Sharp clique packings in almost complete graphs

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Abstract

We determine the maximum number of edges left uncovered by an optimal clique packing in an almost complete graph. For each fixed $r \geq 3$ and every integer sequence $n \ll m \ll n^2$, this maximum over n-vertex graphs with $\binom{n}{2} - m$ edges is $(r-2)n/2 + (c_r + o(1))m^{3/2}/n$, where $c_r = 2\sqrt{2(r-2)}/(3\sqrt{3})$. The fractional maximum is $(c_r + o(1))m^{3/2}/n$. The error is smaller than the structural term throughout this range, including where the degree-divisibility loss is dominant. When $m/n \to \lambda < \infty$, we determine explicit integer and fractional profiles through the $\sqrt{n}$ scale; the integer profile also records the residue of n modulo $r-1$. In the superlinear range, near-extremal graphs have a common neighborhood for their low-degree vertices and are close to a three-part construction. The proof bounds the missing edges forced by these vertices and their uncovered neighborhoods within a single budget. Known decomposition results control the dense core, and a construction with prescribed degree residues attains both losses simultaneously. We also give uniform finite estimates, with a sharper remainder for triangles.

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1 Introduction

A graph with very few missing edges contains a large collection of edge-disjoint cliques. The natural question is how many edges must remain uncovered in the worst case, when only the number of missing edges is specified. High minimum degree permits almost complete packings, and, subject to divisibility conditions, exact decompositions. An average density assumption allows vertices of much smaller degree, however, and these vertices can force a larger loss. We determine the sharp structural loss after separating the obstruction from degree residues, for every fixed clique order.

All graphs in this paper are finite and simple. For an integer $r \geq 3$, let $\nu_r(G)$ be the maximum number of pairwise edge-disjoint copies of K_r in G. A fractional K_r-packing assigns nonnegative weights to the copies of K_r, with total weight at most one on each edge; its maximum total weight is denoted by $\nu_r^*(G)$. Put $R = \binom{r}{2}$ and

$$D_r(G) = e(G) - R\nu_r(G), \qquad D_r^*(G) = e(G) - R\nu_r^*(G).$$

Thus $D_r(G)$ is the smallest possible number of uncovered edges, and $0 \leq D_r^*(G) \leq D_r(G)$. Throughout the paper, r is fixed.

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Yuster [12, Theorem 2.8] proved that, for each fixed r and sufficiently small fixed $\varepsilon > 0$, every n-vertex graph with at least $(1/2 - \varepsilon)n^2$ edges satisfies

$$D_r(G) \le (2r-3)\varepsilon^{6/5}n^2 + o(n^2).$$

The hypothesis of this theorem concerns only the number of edges. Its proof uses decompositions of complete graphs, fractional packings, and the asymptotic equivalence of integer and fractional packings due to Haxell and Rödl [10]. The bound shows that the uncovered proportion is smaller than the missing proportion. It does not determine the correct exponent or leading constant as the density tends to one.

For triangles, the question also forms part of the full density problem posed by Blumenthal, Lidický, Pehova, Pfender, Pikhurko, and Volec [3, Section 5]. They ask for the sharp asymptotic lower bound on the size of a triangle packing at each prescribed edge density. Their results on decompositions into edges and triangles provide bounds throughout the density range. The present paper determines the first nontrivial term at the complete-graph endpoint, and treats the corresponding question for every fixed r.

For $0 \le m \le \binom{n}{2}$, define

$$\psi_r(n,m) = \min\{\nu_r(G) : v(G) = n,\ e(G) = \tbinom{n}{2} - m\},$$

and define $\psi_r^*(n,m)$ by replacing ν_r with ν_r^*. Our main result gives both extremal quantities in a range in which the number of missing edges may vary with n.

Theorem 1.1. *Fix $r \ge 3$ and put $c_r = 2\sqrt{2(r-2)}/(3\sqrt{3})$. For every integer sequence $m = m(n)$ with $n \ll m \ll n^2$,*

$$\psi_r(n,m) = \frac{\binom{n}{2} - m - (r-2)n/2}{\binom{r}{2}} - \left(\frac{c_r}{\binom{r}{2}} + o(1)\right)\frac{m^{3/2}}{n}.$$

For $\psi_r^*(n,m)$, the same formula holds with the term $(r-2)n/2$ omitted.

In the extremal problem, the linear term is forced by the degree residues modulo $r-1$. It dominates when $m \ll n^{4/3}$, whereas the structural term dominates when $m \gg n^{4/3}$. The error is $o(m^{3/2}/n)$ throughout the stated range, even when this scale is smaller than n. The $o(n^2)$ error in general integer–fractional rounding can exceed this scale, so we construct the integer packing directly while controlling the degree residues of its dense core. The required packing bound follows from a finite estimate.

Theorem 1.2. *For each $r \ge 3$, there are constants $q_r, C_r > 0$ and n_r such that every graph G on $n \ge n_r$ vertices with $m = \binom{n}{2} - e(G) \le q_r n^2$ satisfies*

$$D_r(G) \le \frac{r-2}{2}n + c_r\frac{m^{3/2}}{n} + C_r\left(\frac{m^{7/4}}{n^{3/2}} + \frac{m^{3/4}}{\sqrt{n}} + 1\right). \tag{1}$$

The same upper bound holds for $D_r^*(G)$.

For triangles, a maximum matching in each exceptional neighborhood gives more information than the absence of a further clique in its uncovered set. Using this information improves the remainder.

Theorem 1.3. *There are absolute constants $q_0, C > 0$ and n_0 such that every graph G on $n \geq n_0$ vertices with $m = \binom{n}{2} - e(G) \leq q_0 n^2$ satisfies*

$$D_3(G) \leq \frac{n}{2} + \frac{2\sqrt{2}}{3\sqrt{3}} \frac{m^{3/2}}{n} + C\left(\frac{m^2}{n^2} + 1\right). \tag{2}$$

The same upper bound holds for $D_3^*(G)$.

The fractional problem has a different first-order description when the number of missing edges is linear in n. The extremal construction then has a bounded number of exceptional vertices. Retaining the integer nature of this number determines the profile.

Theorem 1.4. *Fix $r \geq 3$ and $\lambda \geq 0$, and put*

$$F_r(\lambda) = \max_{\substack{z \in \mathbb{Z}_{\geq 0} \\ z \leq \lambda}} z\sqrt{2(r-2)(\lambda - z)}.$$

For every sequence of nonnegative integers $m = m(n)$ with $m/n \to \lambda$,

$$\max_{\substack{v(G)=n \\ e(G)=\binom{n}{2}-m}} D_r^*(G) = \big(F_r(\lambda) + o(1)\big)\sqrt{n}.$$

In particular, $F_r(\lambda) = 0$ for $0 \leq \lambda \leq 1$. For $\lambda > 1$, the positive maximizing branches change at $\lambda_j = (3j^2 + 3j + 1)/(2j + 1)$, $j \geq 1$; at λ_j, the values corresponding to $z = j$ and $z = j + 1$ agree. Thus the first transition occurs at $\lambda_1 = 7/3$.

The integer problem has an additional residue-dependent profile. Put $s = r - 1$ and $p = r - 2$. For $a \in \{0, \ldots, p\}$ and $j \geq 0$, set

$$B_{r,a}(u) = \frac{1}{2}\left(a + (p - a)\min\left\{1, \frac{2u}{a+1}\right\}\right), \qquad \tau_{a,j} = j + \frac{(a + 1 - j) \bmod s}{2},$$

where residues lie in $\{0, \ldots, s - 1\}$, and define

$$J_{r,a}(\lambda) = \max\left(\{0\} \cup \left\{j\sqrt{2p(\lambda - \tau_{a,j})} : j \in \mathbb{Z}_{\geq 1},\ \tau_{a,j} \leq \lambda\right\}\right).$$

The thresholds $\tau_{a,j}$ increase strictly with j, so the maximum in this definition is finite.

Theorem 1.5. *Fix $r \geq 3$, $a \in \{0, \ldots, r - 2\}$, and $\lambda \geq 0$. For every sequence of integers $n \equiv a + 1 \pmod{r-1}$ and $m = m(n) \geq 0$ with $m/n \to \lambda$,*

$$\max_{\substack{v(G)=n \\ e(G)=\binom{n}{2}-m}} D_r(G) = nB_{r,a}(m/n) + \big(J_{r,a}(\lambda) + o(1)\big)\sqrt{n}.$$

Moreover, if $a < r - 2$, then uniformly for $0 \leq m \leq (a + 1)n/2$ the same maximum equals $nB_{r,a}(m/n) + O_r(1)$.

The term $nB_{r,a}(m/n)$ comes from optimizing the degree residues. Its argument must retain the actual ratio m/n, because $m/n \to \lambda$ permits $|m - \lambda n|$ to exceed $\sqrt{n}$. The threshold $\tau_{a,j}$ also accounts for the missing edges needed to retain the largest residues after removing j exceptional vertices. For triangles, the linear coefficient is $1/2$ when n is even and $\min\{m/n, 1/2\}$ when n is odd. The first positive square-root branch starts at $\lambda = 3/2$ in the former case and at $\lambda = 1$ in the latter.

The obstruction giving the constant has three vertex sets. The first induces a clique, the second a balanced complete $(r-2)$-partite graph, and the third an independent set. All edges between consecutive sets are present, and there are no edges between the first and third sets. Every edge between the second and third sets lies in no K_r. For triangles, this is the construction used in the study of triangular edges by Füredi and Maleki [6] and Gruslys and Letzter [7]. Gruslys and Letzter determined the maximum number of edges lying in no triangle at each prescribed edge count, for sufficiently large graphs. Our upper bound on the extremal packing number uses the same obstruction. The matching packing bound for arbitrary graphs must also control uncovered edges that do lie in cliques.

In the superlinear range, near equality also forces the structure of the construction. The relevant scales are m/n exceptional vertices and $\sqrt{m}$ vertices in their common neighborhood. The next theorem controls both scales.

Theorem 1.6. *Fix $r \geq 3$, put $p = r-2$, and let $G = G_n$ be a sequence of n-vertex graphs with $n \ll m = \binom{n}{2} - e(G) \ll n^2$. Suppose that $D_r(G) \geq (r-2)n/2 + (c_r - o(1))m^{3/2}/n$. Then there is a partition $V(G) = A \cup B_1 \cup \cdots \cup B_p \cup C$, with $B = \bigcup_i B_i$, such that*

$$|C| = \left(\frac{2}{3} + o(1)\right)\frac{m}{n}, \qquad |B_i| = \left(\frac{1}{p} + o(1)\right)\sqrt{\frac{2pm}{3}}, \qquad \max_{x \in C} |N_G(x) \triangle B| = o(\sqrt{m}).$$

Let H be the graph in which A is a clique, each B_i and C is independent, all pairs between distinct B_i are complete, and the pairs AB and BC are complete while AC is empty. Then $|E(G) \triangle E(H)| = o(m)$. Moreover, only $o(m^{3/2}/n)$ pairs in this symmetric difference have an endpoint in C. For fractional packings, the same conclusions hold in this range under the assumption $D_r^*(G) \geq (c_r - o(1))m^{3/2}/n$.

The proof removes exceptional vertices and then decomposes the remaining graph of high minimum degree. This general strategy, including neighborhood matchings and the final decomposition, appears for triangles in [3, proof of Lemma 3.4]. We use the Hajnal–Szemerédi theorem [9, 11] to handle larger cliques. The dense-graph packing results of Alon, Caro, and Yuster [1] give the sharp linear allowance in the final core. For fractional packings, we use the high-minimum-degree decomposition theorem of Barber, Kühn, Lo, Montgomery, and Osthus [2]. The sharp constant comes from a joint estimate for the missing edges used by the two parts of the argument. Exceptional vertices that have many uncovered incident edges consume missing edges outside the dense core; the uncovered neighborhoods consume missing edges inside it. Optimizing this common budget yields the exponent $3/2$ and the constant c_r. At the linear scale, the number of exceptional vertices is bounded. For integer packings, retaining the core degree residues changes the available missing-edge budget and gives the thresholds $\tau_{a,j}$.

At a smaller scale, Gruslys and Letzter [8, Theorem 1.1] proved that every graph on $n \geq 7$ vertices with at most $n-4$ missing edges has a fractional triangle decomposition. Their exact result concerns a range where the fractional loss is zero. Our estimates concern a larger range and distinguish the degree-divisibility loss in integer packings from the structural obstruction shared by both problems. An exact fractional decomposition of the core removes this linear term. For fractional packings with m/n tending to a finite limit, Theorem 1.4 gives the discrete profile at scale $\sqrt{n}$, rather than an exact decomposition threshold.

Under divisibility assumptions, De Vas Gunasekara and Horsley [4, Theorems 2–3 and Corollary 4] proved exact K_r-decomposition results for graphs with a linear number of missing edges, including sharp thresholds for triangles. Our linear-defect theorems optimize the uncovered-edge count over all graphs with the prescribed number of missing edges, allowing arbitrary degree residues.

Section 2 proves the packing bounds, the matching constructions, the integer and fractional profiles at linear defect, and the sharper triangle estimate. Section 3 proves the stability theorem. Its key step improves a maximum vertex-disjoint clique packing inside an exceptional neighborhood, forcing that entire neighborhood to align with the common multipartite set. Unless otherwise specified, implicit constants in $O_r(\cdot)$ depend only on r; constants without a subscript are absolute when $r = 3$.

2 Packings near complete graphs

We prove the finite packing bound by processing exceptional vertices and then packing the dense core. Matching constructions give the superlinear asymptotic formula. At the linear scale, we retain the number of exceptional vertices and the individual core degree residues; for triangles, a stronger neighborhood estimate improves the finite error. [1]

We first record the known packing results for the core. The following bound is a direct consequence of Alon, Caro, and Yuster [1, Corollary 3.4 and Lemma 3.5].

Lemma 2.1 (Alon–Caro–Yuster)**.** *For each $r \geq 3$, there are $\gamma_r > 0$ and N_r such that every graph J on $N \geq N_r$ vertices with $\delta(J) \geq (1-\gamma_r)N$ satisfies $D_r(J) \leq (r-2)N/2 + \binom{r}{2}$.*

After decreasing γ_r and increasing N_r, every such J also has a fractional K_r-decomposition [2, Theorem 1.3].

The neighborhood step uses the Hajnal–Szemerédi theorem in its clique-factor form. If s divides t and a graph on t vertices has minimum degree at least $(1-1/s)t$, then it has a K_s-factor [9, 11]. In particular, if the maximum degree of its complement is at most D, then deleting at most $s-1$ vertices leaves such a factor whenever the original order is at least $s(D+s+1)$.

Proof of Theorem 1.2. Fix r, put $s = r-1$, and choose $q_r > 0$ sufficiently small in terms of r and Lemma 2.1. All subsequent restrictions on n_r depend only on these choices. When $m = 0$, apply Lemma 2.1 to K_n. We may therefore assume $q = m/n^2 > 0$ and put $\varepsilon = q^{1/4}$.

Let C consist of the vertices whose degree in the complement of G is at most εn, and put $B = V(G) \setminus C$ and $b = |B|$. Separate B into $M = \{x \in B : d_G(x) \geq 0.9n\}$ and $T = B \setminus M$, and put $h = |T|$. The complement degree sum gives

$$b \leq 2q^{3/4}n, \qquad h \leq 40qn \tag{3}$$

for $n \geq 20$, since every vertex of T has complement degree at least $0.05n$. Discard all edges inside B with an endpoint in T; there are at most hb such edges. Every clique selected below is added to the packing and its edges are immediately deleted. Edges explicitly discarded are counted as uncovered.

We first process the vertices of M. The edges between these vertices cannot all be discarded, because b^2 can be of the same order as the leading term. They can instead be covered while using only a bounded number of edges at each core vertex in each stage.

Claim 1. *There is a K_r-packing that covers every remaining edge incident with M, apart from at most $(s-1)|M|$ discarded edges. During its construction, the maximum degree of the graph of deleted edges inside C is at most $(r-2)|M|$. No edge between T and C is used.*

[1] During the research, OpenAI Codex was used to suggest and examine candidate arguments for the missing-edge budget, the linear profiles, and the stability analysis. The mathematical arguments are given in the proofs below, with the external results cited where they are used.

Proof. Process M in any order. When processing x, first consider each of its neighbors y among the as yet unprocessed vertices of M. Complete xy to a K_r using $r-2$ vertices of C, choosing these core vertices disjointly for the different choices of y in this stage.

The original common neighborhood of x, y in C has size at least $0.8n - b$. Each preceding stage removed at most $r-2$ edges from either x or y to C. In the current stage at most $(r-2)b$ core vertices have been reserved. Thus the available common neighborhood has size at least $0.8n - O_r(b)$. Inductively, the maximum complement degree of the current core is at most $\varepsilon n + (r-2)b$. For small enough q_r, one can choose $r-2$ pairwise adjacent vertices greedily from this common neighborhood. This completes each required copy of K_r.

After these choices, the remaining core neighborhood W of x has size at least $0.9n - O_r(b)$. The current graph on W has complement maximum degree at most $D = \varepsilon n + (r-2)b$. Its order exceeds $s(D+s+1)$ when q_r is small and n_r is large. The Hajnal–Szemerédi theorem therefore gives vertex-disjoint copies of K_s covering all but at most $s-1$ vertices of W. Extend each by x, and discard the at most $s-1$ remaining edges from x to C. Remove x from further consideration.

Within this stage, each core vertex belongs to at most one selected clique. Indeed, the core vertices used with edges xy were chosen disjointly, and their edges to x have already been removed before W is considered. A selected clique deletes at most $r-2$ core edges at any core vertex. This proves the inductive bound on deleted core degrees and justifies the choices throughout the procedure. The cliques used contain no vertex of T, which proves the last assertion. □

Process the vertices of T next. For $x \in T$, let $W_x = N_G(x) \cap C$. These edges from x to C are still present when x is reached. Take a maximum collection of vertex-disjoint copies of K_s in the current graph on W_x, extend them by x, and discard the edges from x to the uncovered vertices U_x. The set U_x spans no K_s in the current core. As in Claim 1, each stage deletes at most $r-2$ core edges at a core vertex. Consequently, at every time the maximum complement degree of the current core is at most

$$D = \varepsilon n + (r-2)b. \tag{4}$$

If $|W_x| \geq s(D+s+1)$, the clique-factor theorem ensures $|U_x| \leq s-1$. Let Z be the vertices of T for which this size condition fails, and put $z = |Z|$ and $\ell = z/n$. For a constant K depending only on r,

$$z \leq 40qn, \qquad d_G(x) \leq K(\varepsilon n + 1) \quad (x \in Z). \tag{5}$$

The degree estimate follows from $d_G(x) \leq |W_x| + b$, (3), and (4).

Let m_C be the number of missing edges in the original graph $G[C]$. We now use a single budget for m_C and the missing edges forced by Z. There are at least $z(n-z) - \sum_{x \in Z} d_G(x)$ missing edges between Z and its complement, and these are disjoint from the missing edges inside C. By (5),

$$\frac{m_C}{n^2} \leq Q - \ell, \qquad Q = q + \ell^2 + K(\varepsilon + n^{-1})\ell = q + O_r(\varepsilon q + q/n). \tag{6}$$

In particular, $0 \leq \ell \leq Q$.

For $x \in Z$, write $u = |U_x|$. Turán's theorem shows that a K_s-free graph on u vertices is missing at least $u^2/(2(s-1)) - u/2$ edges. The missing edges in the current core on U_x consist of at most m_C original missing edges and at most $(r-2)bu/2$ edges used in preceding cliques. Hence

$$\frac{u^2}{2(r-2)} - \frac{u}{2} \leq m_C + \frac{(r-2)bu}{2}, \qquad u \leq \sqrt{2(r-2)m_C} + O_r(b+1),$$

where the second inequality follows by solving the quadratic inequality.

The remaining graph on C has minimum degree at least $|C| - 1 - \varepsilon n - (r-2)b$. Our choices of q_r and n_r ensure that Lemma 2.1 applies to it, leaving at most $(r-2)|C|/2 + \binom{r}{2}$ additional uncovered edges. The selected cliques and this final packing are edge-disjoint. The uncovered edges therefore number at most

$$\begin{aligned} D_r(G) &\le \frac{r-2}{2}n + z\sqrt{2(r-2)m_C} + O_r(hb + zb + b + 1) \\ &\le \frac{r-2}{2}n + n^2\ell\sqrt{2(r-2)(Q-\ell)} + O_r(q^{7/4}n^2 + q^{3/4}n + 1). \end{aligned} \tag{7}$$

The error includes the initial hb discarded edges, the at most $s-1$ leftover edges for each vertex outside Z, the error $O_r(b+1)$ for each vertex in Z, and the constant remainder from the core. Here $h, z \le b$, and the linear part of the core loss has been displayed separately. The estimates in (3) justify the second line.

For $0 \le \ell \le Q$, the function $\ell\sqrt{2(r-2)(Q-\ell)}$ has its maximum at $\ell = 2Q/3$, where its value is $c_r Q^{3/2}$. By (6), $Q^{3/2} = q^{3/2} + O_r(q^{7/4} + q^{3/2}/n)$. Since $q \le 1$, the term $q^{3/2}n$ is at most $q^{3/4}n$. Substituting in (7) yields $D_r(G) \le (r-2)n/2 + c_r q^{3/2}n^2 + O_r(q^{7/4}n^2 + q^{3/4}n + 1)$, which is (1). Since $D_r^*(G) \le D_r(G)$, the fractional assertion follows as well. □

For fractional packings, the final core can be decomposed exactly. Thus the same argument gives the following estimate without a linear allowance for degree residues.

Proposition 2.2. *For each fixed $r \ge 3$, there are constants $C_r', q_r' > 0$ such that, for all sufficiently large n, every graph with $m = \binom{n}{2} - e(G) \le q_r' n^2$ satisfies*

$$D_r^*(G) \le c_r \frac{m^{3/2}}{n} + C_r' \left(\frac{m^{7/4}}{n^{3/2}} + \frac{m^{3/4}}{\sqrt{n}} + 1 \right). \tag{8}$$

For $r = 3$, the error may be replaced by $C_3'(m^2/n^2 + 1)$.

Proof. Use the exceptional-vertex construction in the proof of Theorem 1.2, replacing the final integer packing by a fractional decomposition of the core, supplied by [2, Theorem 1.3]. Give weight one to the previously selected cliques; their edges are disjoint from the core that remains. The estimate (7) therefore holds for $D_r^*(G)$ with the term $(r-2)n/2$ omitted. The same optimization proves (8). For triangles, make the identical replacement in the proof of Theorem 1.3 below. Only the final core contribution changes, giving the stated error. □

We next give the matching construction, including the adjustment to an arbitrary prescribed integer number of missing edges. This adjustment is needed because the parameter m in Theorem 1.1 is not restricted to a subsequence.

Proposition 2.3. *For each fixed $r \ge 3$, there are constants $q_r', A_r, C_r' > 0$ and n_r' such that, whenever $n \ge n_r'$ and $A_r n \le m \le q_r' n^2$ is an integer, there is an n-vertex graph with exactly $\binom{n}{2} - m$ edges and*

$$D_r^*(G) \ge c_r \frac{m^{3/2}}{n} - C_r' \left(\frac{m^2}{n^2} + \sqrt{m} \right). \tag{9}$$

The same lower bound holds for $D_r(G)$.

Proof. Put $q = m/n^2$ and choose positive real numbers

$$\beta = \sqrt{\frac{2(r-2)q}{3}}, \qquad \gamma = 1 - \beta - \sqrt{(1-\beta)^2 - \frac{4q}{3}}, \qquad \alpha = 1 - \beta - \gamma.$$

For sufficiently small q_r', these are well-defined and satisfy $\alpha > 0$, $\gamma = 2q/3 + O_r(q^{3/2})$, and

$$q = \gamma(1-\beta) - \frac{\gamma^2}{2} + \frac{\beta^2}{2(r-2)}, \qquad \beta\gamma = c_r q^{3/2} + O_r(q^2). \tag{10}$$

Partition n vertices into A, B, C with sizes within $O_r(1)$ of $\alpha n, \beta n, \gamma n$, respectively. Let A induce a clique, B a balanced complete $(r-2)$-partite graph, and C an independent set. Include all edges between A, B and between B, C, and no edges between A, C. When $r = 3$, the graph on B is independent.

No edge between B and C belongs to a K_r. A clique containing a vertex of C would need its other $r-1$ vertices in B, whose clique number is at most $r-2$. By (10), the graph just constructed has $m + O_r(n)$ missing edges and

$$e(B,C) = c_r \frac{m^{3/2}}{n} + O_r\left(\frac{m^2}{n^2} + \sqrt{m}\right). \tag{11}$$

For the count of missing edges, the three contributions are $|A||C|$, $\binom{|C|}{2}$, and the edges missing within the $r-2$ parts of B; their quadratic terms are precisely the first identity in (10). For the edge count $e(B,C)$, rounding changes the product $\beta\gamma n^2$ by only $O_r(\beta n + \gamma n + 1) = O_r(\sqrt{m})$.

If the number of missing edges exceeds m, move vertices one at a time from C to A, retaining the same construction rule, until the number is at most m. Such a move increases the edge count by the previous size of A, which is at least $n/2$ when q_r' is small. At most $O_r(1)$ moves are needed. Taking A_r sufficiently large ensures that there are enough vertices of C, since $|C| = 2m/(3n) + O_r(q^{3/2}n + 1)$. The remaining difference from m is $O_r(n)$, and can be made up by deleting edges inside A. There are $\Theta(n^2)$ such edges, so this last step is possible for large n.

The moves change $e(B,C)$ by at most $O_r(\sqrt{m})$, and the final deletions do not change any edge between B and C. These edges still belong to no K_r. In every fractional packing, the total covered edge weight is at most $e(G) - e(B,C)$, and therefore $R\nu_r^*(G) \le e(G) - e(B,C)$. Equation (11) proves (9). The integer assertion follows from $D_r(G) \ge D_r^*(G)$. □

For integer packings, the structural obstruction can coexist with the largest degree-residue obstruction. The adjustment below reserves only $O_r(n)$ missing edges, which costs $O_r(\sqrt{m})$ in the structural term.

Proposition 2.4. *Fix $r \ge 3$. For every integer sequence $n \ll m \ll n^2$, there is a graph G on n vertices with exactly $\binom{n}{2} - m$ edges such that*

$$D_r(G) \ge \frac{r-2}{2} n + c_r \frac{m^{3/2}}{n} - O_r\left(\frac{m^2}{n^2} + \sqrt{m}\right).$$

Proof. Put $s = r-1$ and $p = r-2$, and fix a sufficiently large integer $K = K(r)$. Apply the rounded three-part construction in the proof of Proposition 2.3 with target $m' = m - Kn$, before moving any vertices or deleting any edges inside A. This is possible for all sufficiently large n, since $m/n \to \infty$. The raw construction has $m_0 = m' + O_r(n)$ missing edges, and its part A is a clique of order $a = n - O_r(\sqrt{m} + m/n) \sim n$. The constant in $m_0 - m' = O_r(n)$ is independent of K. Choose K large enough that $T = m - m_0$ satisfies $sn \le T \le C_r n$ for a fixed C_r.

Every vertex in A initially has degree $d_0 = n - |C| - 1$. Let $\delta \in \{0, \dots, s-1\}$ be congruent to $d_0 - p$ modulo s. Start with the prescribed degree $s + \delta$ at every vertex of A, and increase one entry by some $t \in \{0, \dots, s-1\}$ so that their sum S is congruent to $2T$ modulo s. For sufficiently large

n, $S \le 2T$, because $S \le (2s-1)a + s - 1$ and $T \ge sn$. Distribute $(2T - S)/s$ increments of size s among the a entries as evenly as possible. The resulting positive integers f_v have sum $2T$, are bounded by a constant depending only on r, and satisfy $f_v \equiv d_0 - p \pmod{s}$ at all but at most one vertex. The prescribed-degree lemma of Alon, Caro, and Yuster [1, Lemma 2.2], applied to the complete graph on A, gives a graph $F \subseteq G[A]$ with degrees f_v. Its hypotheses hold since the degrees are positive and bounded, their sum is even, and $a \to \infty$.

Delete the T edges of F. The resulting graph has exactly m missing edges, and all but at most one vertex of A now have degree congruent to $p = s - 1$ modulo s. In the leave of any K_r-packing, each such vertex has degree at least p, since deleting a copy of K_r changes its degree by s. Also, every edge between B and C must remain uncovered. These edges have no endpoint in A, so the two requirements give

$$D_r(G) \ge |B||C| + \frac{p(a-1)}{2}.$$

The construction estimate at m' gives $|B||C| = c_r(m')^{3/2}/n + O_r(m^2/n^2 + \sqrt{m})$. Moreover, $|m^{3/2} - (m')^{3/2}|/n = O_r(\sqrt{m})$, and $n - a = O_r(\sqrt{m} + m/n) = O_r(\sqrt{m})$. Substitution proves the asserted bound. □

Proof of Theorem 1.1. For integer packings, apply Theorem 1.2 and Proposition 2.4. If $n \ll m \ll n^2$, the ratios of $m^{7/4}/n^{3/2}$, $m^{3/4}/\sqrt{n}$, m^2/n^2, and $\sqrt{m}$ to $m^{3/2}/n$ are, respectively, $(m/n^2)^{1/4}$, $\sqrt{n}/m^{3/4}$, $\sqrt{m}/n$, and n/m. All tend to zero, as does $n/m^{3/2}$, the ratio for the constant error. Thus the largest possible value of $D_r(G)$ is $(r-2)n/2 + (c_r + o(1))m^{3/2}/n$.

For fractional packings, use Proposition 2.2 and Proposition 2.3. The same error comparisons show that the largest fractional loss is $(c_r + o(1))m^{3/2}/n$. The identities defining D_r and D_r^* give the asserted packing formulas. □

We next consider a linear number of missing edges. The number of low-degree exceptional vertices is now bounded, so the continuous optimization in the finite bound must retain this integer parameter. An exact fractional decomposition of the dense core leaves only the exceptional neighborhoods to be counted.

Proof of Theorem 1.4. Put $p = r - 2$ and repeat the exceptional-vertex construction in the proof of Theorem 1.2, taking $\varepsilon = n^{-1/4}$. In the present proof, all implicit constants may depend on r and λ. Denote the original core by K and the exceptional set by E, and put $b = |E|$. Then $b \le 2m/(\varepsilon n) = O(n^{1/4})$. The number h of exceptional vertices of original degree less than $0.9n$ is $O(1)$, since each has complement degree at least $0.05n$ for large n. The high-degree exceptional vertices can be processed by Claim 1: its greedy choices and clique factors apply because $\varepsilon = o(1)$ and $b = o(n)$. Then process the remaining exceptional vertices by maximum vertex-disjoint K_{r-1}-packings in their current core neighborhoods, as before. During these stages, the used core edges have maximum degree at most pb.

Let Z be the small-neighborhood set from that construction, $z = |Z|$, and let M be the number of missing edges in the original graph $G[K]$. For $x \in Z$, write u_x for the number of vertices left uncovered in its core neighborhood. Here $z \le h = O(1)$ and $d_G(x) = O(\varepsilon n + b + 1) = O(n^{3/4})$ for $x \in Z$. The shared missing-edge budget and Turán's theorem therefore give

$$M \le m - zn + O(n^{3/4}), \qquad u_x \le \sqrt{2pM} + O(b+1).$$

For the second bound, the current graph on the uncovered set is K_{p+1}-free, so $u_x^2/(2p) - u_x/2 \le M + pbu_x/2$.

The remaining core has minimum degree at least $|K| - 1 - \varepsilon n - pb = (1 - o(1))|K|$. Use the fractional decomposition theorem [2, Theorem 1.3] on this core, together with weight one on the cliques already selected. Their edge supports are disjoint. All discarded edges outside the sets counted by u_x number $O(hb + b)$, so

$$D_r^*(G) \le \sum_{x \in Z} u_x + O(hb + b) \le z\sqrt{2pM} + O(n^{1/4}).$$

Since $M \ge 0$, the budget gives $z \le \lambda + o(1)$. The integer z is bounded. On any subsequence where it is constant, $z \le \lambda$ and $M/n \le \lambda - z + o(1)$, yielding $\limsup D_r^*(G)/\sqrt{n} \le z\sqrt{2p(\lambda - z)} \le F_r(\lambda)$. Every subsequence has a further subsequence of this kind, which proves the upper bound.

For the lower bound, there is nothing to prove if $\lambda \le 1$, as the asserted coefficient is zero. Suppose $\lambda > 1$ and choose a maximizing integer j. Then $1 \le j < \lambda$. Use the three-part construction with $|C| = j$, with B a balanced complete p-partite graph of order b, and with $|A| = n - b - j$. The number of missing edges is

$$\begin{aligned} M_0(b) &= j(n - j - b) + \binom{j}{2} + \sum_{i=1}^{p} \binom{|B_i|}{2} \\ &= jn + \frac{b^2}{2p} - \left(j + \frac{1}{2}\right) b - \frac{j^2 + j}{2} + O_r(1). \end{aligned}$$

Choose the largest integer $b \le n - j$ with $M_0(b) \le m$. Since $m - jn \sim (\lambda - j)n > 0$, such a choice exists, and the displayed formula gives $b \sim \sqrt{2p(\lambda - j)n}$. Furthermore, $M_0(b+1) - M_0(b) = \lfloor b/p \rfloor - j = O(\sqrt{n})$, so $0 \le m - M_0(b) = O(\sqrt{n})$. Delete exactly $m - M_0(b)$ additional edges inside A. Every edge between B and C still lies in no K_r, and hence

$$D_r^*(G) \ge jb = \big(F_r(\lambda) + o(1)\big)\sqrt{n}.$$

This completes the proof. The stated transition points follow by equating the squares of the j and $j + 1$ branches. Their crossings are strictly increasing, so the maximizing branches occur in order. □

For integer packings at the linear scale, the dense core also contributes a linear loss. We first estimate that loss from the individual degree residues; the square-root term then requires a joint budget for these residues and the exceptional neighborhoods.

For a graph G, write $\rho_r(G) = \frac{1}{2}\sum_{v \in V(G)} (d_G(v) \bmod (r-1))$, where each residue is chosen in $\{0, \dots, r-2\}$.

Proposition 2.5. *For each $r \ge 3$ there are $q_r > 0$ and n_r such that, whenever $n \ge n_r$ and $0 \le m = \binom{n}{2} - e(G) \le q_r n^2$,*

$$\rho_r(G) \le D_r(G) \le \rho_r(G) + O_r\left(\sqrt{m} + \frac{m^{3/2}}{n} + 1\right).$$

Proof. Put $s = r - 1$ and $p = r - 2$. The lower bound follows because each copy of K_r covers s edges at every one of its vertices. Suppose first that $m > 0$. Repeat the exceptional-vertex construction from the proof of Theorem 1.2, now taking $C = \{v : d_{\overline{G}}(v) \le \sqrt{m}\}$. Then $b = |V(G) \setminus C| \le 2\sqrt{m}$ and the number h of exceptional vertices of degree less than $0.9n$ is at most $40m/n$. Claim 1 applies with εn replaced by $\sqrt{m}$: the maximum complement degree of the current core is at most $\sqrt{m} + pb$, which is sufficiently small compared with n when q_r is small. At each of the remaining h vertices,

the same Turán estimate gives at most $\sqrt{2pm} + O_r(b+1)$ uncovered incident edges. Consequently, the construction discards a total of

$$L = O_r\left(hb + b + h(\sqrt{m} + b + 1)\right) = O_r\left(\sqrt{m} + \frac{m^{3/2}}{n} + 1\right)$$

edges. Its remaining core H has order $N = n - b$ and $\Delta(\overline{H}) \le \Delta := \sqrt{m} + pb$.

Write $\beta_v = d_G(v) \bmod s$ and $\beta'_v = d_H(v) \bmod s$. Every edge removed from the core belongs either to a selected K_r or to a discarded edge. Thus $\beta'_v = \beta_v$ unless v is incident with a discarded edge. Each discarded edge has at most one endpoint in C, so $\sum_{v\in C} \beta'_v \le \sum_{v\in V(G)} \beta_v + pL$. We next realize the residues of H by deleting a sparse graph. This uses the prescribed-degree lemma of Alon, Caro, and Yuster [1, Lemma 2.2].

Let $X = \{v \in C : \beta'_v > 0\}$, put $D_0 = rs$ and $d_0 = \lceil D_0/2 \rceil$, and choose an integer $k_0 \ge (4d_0+2)(\Delta + 2d_0 + 2)$ with $k_0 = O_r(\Delta + 1)$. Our choices of q_r and n_r ensure $k_0 \le N$. If $|X| \ge k_0$, let $Y = X$; otherwise extend X to a set Y of size k_0. Prescribe degree β'_v at every $v \in X$ and degree s at every $v \in Y \setminus X$. The sum S of these prescribed degrees satisfies $S \equiv 2e(H) \pmod s$. Choose $j \in \{0, \dots, r-1\}$ such that $S + sj \equiv 2e(H) \pmod{r(r-1)}$, and increase one prescribed degree by sj. The resulting degrees are positive, at most D_0, and have even sum. Also, $\delta(H[Y]) \ge |Y| - 1 - \Delta \ge |Y| - |Y|/(4d_0+2) + 2d_0 + 1$. The cited lemma therefore supplies a subgraph $F \subseteq H[Y]$ with these degrees. In particular,

$$e(F) = \frac{1}{2}\sum_{v\in C} \beta'_v + O_r(\sqrt{m} + 1), \qquad \Delta(F) \le D_0.$$

The graph $H - F$ is K_r-divisible and has sufficiently high minimum degree, so the dense decomposition theorem [1, Lemma 2.1] decomposes it into copies of K_r. Together with the previously chosen cliques this leaves at most $L + e(F) \le \rho_r(G) + (1 + p/2)L + O_r(\sqrt{m} + 1)$ edges. If $m = 0$, apply just the prescribed-degree argument to $H = K_n$ with $\Delta = 0$; the error is then $O_r(1)$. □

Proof of Theorem 1.5. Put $s = r - 1$ and $p = r - 2$. All implicit constants may depend on r and on a fixed upper bound for m/n. We first optimize the degree residues, including the exact number of missing edges.

Claim 2. *Uniformly for $m = O(n)$, every admissible graph satisfies $\rho_r(G) \le nB_{r,a}(m/n)$, and there is an admissible graph with $\rho_r(G) \ge nB_{r,a}(m/n) - O_r(1)$.*

Proof. Write $k = a + 1$. For every integer $h \ge 0$, $(a - h) \bmod s \le p$ and $(a - h) \bmod s \le a + (p - a)h/(a+1)$. Indeed, the residue equals $a - h$ for $h \le a$ and is at most p otherwise. Applying these inequalities to the complement degrees, whose sum is $2m$, gives the upper bound.

Suppose first that $2m \le kn$. Choose the largest integer $u \le \min\{n, 2m/k\}$ for which ku is even. Then $u = \min\{n, 2m/k\} + O_r(1)$. If $u > k$, take a simple k-regular graph on u vertices and add $n - u$ isolated vertices. The regular graph can be obtained by joining consecutive distances on a cyclic ordering, and, for odd k, adding the antipodal perfect matching; in that case u is even. Add the remaining $m - ku/2 = O_r(1)$ edges arbitrarily and take the complement. Before these last additions, u vertices have degree residue p and the others have residue a. The last additions change the residue sum by $O_r(1)$. If $u \le k$, then $m = O_r(1)$ and arbitrary deletion of m edges from K_n gives the same conclusion.

If $2m \ge kn$, choose $t \in \{0, \dots, s-1\}$ with $t \equiv 2m - kn \pmod s$ and distribute $(2m - kn - t)/s$ units as evenly as possible among the vertices. Prescribe complement degree k plus s times the number of units at each vertex, and increase one prescribed degree by t. The degrees are positive

and bounded, and sum to exactly $2m$. By [1, Lemma 2.2], applied to K_n, they are the degrees of a simple graph for all sufficiently large n. Its complement has degree residue p at all but possibly one vertex, which proves the lower bound. □

Proposition 2.5 and the claim already show that the maximum loss is $nB_{r,a}(m/n) + O(\sqrt{n})$. We next sharpen this on the unsaturated part of the profile. Suppose $a < p$ and $m \le (a+1)n/2$, and write $U = an/2 + (p-a)m/(a+1)$. Choose a graph G maximizing $D_r(G)$ at the prescribed edge count. The claim and Proposition 2.5 imply

$$0 \le 2U - \sum_v \beta_v = O_r(\sqrt{m} + 1), \qquad \beta_v = d_G(v) \bmod s.$$

Put $c = (p-a)/(a+1) > 0$. For $h \ge 0$, the nonnegative quantity $g(h) = a + ch - ((a-h) \bmod s)$ is at least $c(h-a-1)$. Its sum over the complement degrees is the left-hand side above. Thus $\Delta(\overline{G}) = O_r(\sqrt{m}+1)$. If $a \ge 1$, the set $X = \{v : \beta_v > 0\}$ has order $\Omega_r(n)$. If $a = 0$ and m is sufficiently large, it has order $\Omega_r(m)$. In either case its order is large enough compared with $\Delta(\overline{G})$ to apply [1, Lemma 2.2] directly in $G[X]$ to the positive residues, increasing one by at most $s(r-1)$ to meet the edge-count congruence. No auxiliary vertices are needed in the prescribed-degree argument from Proposition 2.5. It leaves at most $\rho_r(G) + O_r(1) \le U + O_r(1)$ edges. When $a = 0$ and m is bounded, the proposition itself gives an $O_r(1)$ error. This proves the last assertion of the theorem.

It remains to determine the square-root term on the saturated part. Let $k_j = (a+1-j) \bmod s$ and $\tau_j = j + k_j/2$. We may suppose that $m \ge \tau_0 n$ and that G is extremal. Indeed, for $a < p$ the complementary case was just settled, while $\tau_0 = 0$ for $a = p$. Claim 2 gives $D_r(G) \ge pn/2 - O_r(1)$. Repeat the construction from the proof of Theorem 1.4, with $\varepsilon = n^{-1/4}$. Use its notation K, E, b, Z, and put $j = |Z|$. Thus $b = O(n^{1/4})$, $j = O(1)$, each $x \in Z$ has original degree $O(n^{3/4})$, and the uncovered sets U_x have sizes $u_x = O(\sqrt{n})$. All discarded edges other than those counted by the u_x number $o(\sqrt{n})$.

Let H be the remaining core and put $\beta'_v = d_H(v) \bmod s$. If fewer than $|K|/(10s^3)$ vertices have positive β'_v, then [1, Lemma 3.5] gives $D_r(H) \le |K|/(5s^2)$. Together with the $O(\sqrt{n})$ discarded edges, this contradicts $D_r(G) \ge pn/2 - O_r(1)$. Hence [1, Corollary 3.4] applies, giving $D_r(H) = \frac{1}{2}\sum_{v\in K} \beta'_v + O_r(1)$. Writing $\mu = \sum_{v\in K}(p - \beta'_v)$, we obtain

$$D_r(G) \le \frac{p(n-b)}{2} - \frac{\mu}{2} + \sum_{x\in Z} u_x + o(\sqrt{n}).$$

It follows that $\mu = O(\sqrt{n})$. Apart from $O(\sqrt{n})$ core vertices, therefore, $\beta'_v = p$. Only $O(\sqrt{n})$ core vertices are incident with discarded edges, so the original degree residue is also p outside $O(\sqrt{n})$ core vertices.

Consider the original complement on $V(G) \setminus Z$, denoted by Q. The vertices of Z have only $O(n^{3/4}) = o(n)$ neighbors in total. Consequently, all but $o(n)$ vertices outside Z have original degree residue p and no neighbor in Z. At every such vertex, $d_Q(v) \equiv n - j - 1 - p \equiv k_j \pmod s$, so $d_Q(v) \ge k_j$. Fix $x \in Z$ and put $U = U_x$ and $u = |U|$. The graph on U when x was processed was K_{p+1}-free, and the edges already used inside the core have maximum degree at most $pb = O(n^{1/4})$. Thus

$$e(Q[U]) \ge \frac{u^2}{2p} - \frac{u}{2} - \frac{pbu}{2} = \frac{u^2}{2p} - o(n).$$

To include the residue cost without double counting, give full weight to edges of $Q[U]$ and half the complement degree to each vertex outside $Z \cup U$. No edge of Q is counted more than once, giving

$$e(Q) \ge e(Q[U]) + \frac{1}{2}\sum_{v\notin Z\cup U} d_Q(v) \ge \frac{u^2}{2p} + \frac{k_j n}{2} - o(n).$$

There are also at least $jn - o(n)$ missing edges with an endpoint in Z, disjoint from those counted in Q. Therefore $m \geq \tau_j n + u_x^2/(2p) - o(n)$ for every $x \in Z$. On any subsequence where the bounded integer $j \geq 1$ is constant, this implies $\tau_j \leq \lambda$ and

$$\limsup \frac{D_r(G) - pn/2}{\sqrt{n}} \leq j\sqrt{2p(\lambda - \tau_j)} \leq J_{r,a}(\lambda).$$

When $j = 0$, the same upper bound follows directly with right-hand side zero. Passing to such subsequences proves the upper bound. At $\lambda = \tau_0$, the previously treated unsaturated subsequences must retain the center $nB_{r,a}(m/n)$; on the saturated subsequences this center is exactly $pn/2$.

For the matching lower bound, the claim suffices if $J_{r,a}(\lambda) = 0$. Otherwise choose a maximizing $j \geq 1$, so $\lambda > \tau_j$, and put $k = k_j$ and $N = n - j$. Use the three-part construction with $|C| = j$, with B balanced complete p-partite of order b, and with $|A| = N - b$. Let $M_0(b)$ be its number of missing edges. As before,

$$M_0(b) = jn + \frac{b^2}{2p} - \left(j + \frac{1}{2}\right) b - \frac{j^2 + j}{2} + O_r(1), \qquad M_0(b+1) - M_0(b) = \lfloor b/p \rfloor - j.$$

Choose a sufficiently large fixed integer $K_0 = K_0(r)$, for example $K_0 = 4s^2 + 4s$, and take the largest integer $b \leq N$ for which $M_0(b) + k(N-b)/2 + K_0 b \leq m$. Since $\lambda > \tau_j$, this choice exists for large n and satisfies

$$b = (\sqrt{2p(\lambda - \tau_j)} + o(1))\sqrt{n}, \qquad T := m - M_0(b) = \frac{k|A|}{2} + K_0 b + O_{r,j}(b+1).$$

In particular, $2T \geq k|A| + sb$. Write H_0 for the induced graph on $A \cup B$. We delete exactly T edges from H_0 so that all but at most one of its remaining vertex degrees have residue p modulo s.

At each vertex in a part B_i, prescribe the unique degree $f_v \in \{1, \ldots, s\}$ with $f_v \equiv N - |B_i| - p \pmod{s}$, and put $S_B = \sum_{v \in B} f_v \leq sb$. If $k > 0$, also prescribe degree k at every vertex of A. Let $t \in \{0, \ldots, s-1\}$ be the residue of $2T - k|A| - S_B$ modulo s. Increase the prescribed degree by s at $(2T - k|A| - S_B - t)/s = O_{r,j}(b)$ distinct vertices of A, and by t at one vertex of A. These positive degrees are at most $3s - 2$ and sum to exactly $2T$. Since $\Delta(\overline{H_0}) = O(b) = o(n)$, [1, Lemma 2.2] realizes them by a subgraph of H_0.

If $k = 0$, put $u = \lfloor (2T - S_B)/s \rfloor$ and $t = 2T - S_B - su$. Choose u auxiliary vertices in A, prescribe degree s at each, and increase one of these degrees by t. All other vertices of A are excluded from the prescribed-degree host. Here $u = O_{r,j}(b) = o(n)$ and $u \geq (2K_0/s - 1)b - 1$. Hence the host consisting of B and these auxiliary vertices has order sufficiently large compared with b, whereas its complement has maximum degree at most b. The positive prescribed degrees are at most $2s - 1$ and have even sum $2T$, so [1, Lemma 2.2], with parameter s, realizes them in this host as well.

In either case let F be the realized graph and delete it from the three-part construction. The resulting graph G has exactly m missing edges. Its induced subgraph $H = H_0 - F$ has degree residue p at all but at most one vertex. Every edge between B and C belongs to no K_r, and in fact all copies of K_r lie in H. Consequently,

$$D_r(G) = jb + D_r(H) \geq jb + \rho_r(H) \geq \frac{pn}{2} + jb - O_{r,j}(1).$$

The value of b gives the required lower bound, completing the proof. □

For triangles, a fixed threshold suffices to select the core. This makes the number of exceptional vertices $O(qn)$, where $q = m/n^2$, and permits us to discard all edges between them at cost $O(q^2 n^2)$.

The additional ingredient is the absence of a short augmenting path for a maximum matching. It relates the number of uncovered vertices to the size of the whole neighborhood, rather than only to the number of missing edges in its uncovered set.

Proof of Theorem 1.3. Fix a sufficiently small constant $\eta > 0$ in terms of Lemma 2.1 with $r = 3$, and then choose $q_0 > 0$ sufficiently small in terms of η. All lower bounds on n are absorbed into n_0. The case $m = 0$ follows from that lemma, so assume $q = m/n^2 > 0$.

Let C consist of the vertices whose complement degree in G is at most ηn, and put $B = V(G) \setminus C$, $b = |B|$, and $N = |C|$. Then $b \le 2qn/\eta = O(qn)$. Let h be the number of missing edges in the original graph $G[C]$, and put $S = \sqrt{2h}$. Discard all edges inside B, of which there are at most $b^2/2$.

Process B in any order. For each $x \in B$, take a maximum matching in its neighborhood $W = N_G(x) \cap C$ in the current core. Add the triangles obtained by adjoining x, delete their edges, and discard the remaining edges from x to C. At each stage, a core vertex loses at most one core edge. Thus the graph of deleted core edges has maximum degree at most b throughout. The edges from x to C have not been affected by the preceding stages, so W is its original core neighborhood.

Let $M = \{x \in B : d_G(x) \ge 0.9n\}$ and $T = B \setminus M$. For $x \in M$, the order of W is at least $0.9n - b$, and its complement maximum degree is at most $\eta n + b$. The clique-factor consequence stated above, with $s = 2$, shows that the matching leaves at most one vertex uncovered. It remains to bound the leftover edges for vertices of T.

Claim 3. *For $x \in T$, let u_x be the number of vertices left uncovered by the maximum matching used when x is processed. Then*

$$u_x \le \frac{S}{N - S}\bigl(n - d_G(x)\bigr) + b + 1. \tag{12}$$

Proof. Write $t = |W|$ and $u = u_x$. If $u \le 1$, the assertion holds, so suppose $u \ge 2$. The maximum-matching estimate in [3, p. 278, equation (3.3)] gives at least $\binom{u}{2} + u(t-u)/2 = u(t-1)/2$ missing edges in the current graph on W. At most h were missing originally and at most $bt/2$ have been deleted during previous stages. It follows that

$$u(t-1) \le S^2 + bt. \tag{13}$$

Set $v = \max\{u - b - 1, 0\}$. Since $u \le t$, equation (13) gives $vt \le S^2$, and hence $v \le \min\{t, S^2/t\}$ when $t > 0$. The case $t = 0$ has $v = 0$.

Our choice of q_0 ensures $S \le N/2$ and $0.9n \le N - S$. Since $x \in T$, it follows that $0 \le t \le N - S$. For this range,

$$\min\{t, S^2/t\} \le \frac{S}{N - S}(N - t),$$

with value zero used when $t = 0$ or $S = 0$. For $t \le S$, the inequality follows from $t/(N-t) \le S/(N-S)$. For $S \le t \le N - S$, it follows from $t(N-t) \ge S(N-S)$. Finally, $d_G(x) \le t + b$ implies $N - t \le n - d_G(x)$. Adding back $b + 1$ proves (12). □

Put $a = S/(N-S)$ and $\rho = |T|$. The complement degree sum over T counts a missing edge inside T twice and any other incident missing edge once. No such edge is internal to C. Therefore

$$\sum_{x \in T}\bigl(n - d_G(x)\bigr) \le m - h + \binom{\rho}{2} + \rho. \tag{14}$$

The last ρ accounts for the difference between $n - d_G(x)$ and the complement degree $n - 1 - d_G(x)$.

After all vertices of B are processed, the remaining core has minimum degree at least $N - 1 - \eta n - b$. By our choices of η, q_0 and n_0, Lemma 2.1 supplies a triangle packing of this core leaving at most

$N/2+3$ edges. The initial discarded edges, the at most $|M|$ leftovers from M, and the bounds in (12) and (14) give

$$\begin{aligned} D_3(G) &\le \frac{b^2}{2} + |M| + a\left(m - h + \binom{\rho}{2} + \rho\right) + \rho(b+1) + \frac{N}{2} + 3 \\ &\le \frac{n}{2} + a(m-h) + O(q^2n^2+1). \end{aligned}$$

Here $b, \rho = O(qn)$ and $a = O(\sqrt{q})$; all constants are absolute because η has been fixed; the remaining $O(qn)$ terms are absorbed using $qn \le q^2n^2+1$.

Write $y = h/n^2$, so $0 \le y \le q$. Since $b/n = O(q)$,

$$a = \frac{\sqrt{2y}}{1 - b/n - \sqrt{2y}} = \sqrt{2y} + O(q).$$

Consequently $D_3(G) \le n/2 + n^2(q-y)\sqrt{2y} + O(q^2n^2+1)$. The function $(q-y)\sqrt{2y}$ has its maximum on $[0,q]$ at $y = q/3$, where it equals $2\sqrt{2}\,q^{3/2}/(3\sqrt{3})$. This proves (2), and $D_3^*(G) \le D_3(G)$ gives the fractional assertion. □

For completeness, we translate this estimate into the normalization of the density problem in [3, Section 5]. For $d \in [0,1]$, put

$$f(d) = \liminf_{n\to\infty} \min_{\substack{v(G)=n \\ e(G)=\lfloor d\binom{n}{2}\rfloor}} \frac{3\nu_3(G)}{\binom{n}{2}}.$$

Define $f^*(d)$ with ν_3^* in place of ν_3. The use of a lower limit makes no assumption about the existence of the full density limit.

Corollary 2.6. *As $d \uparrow 1$,*

$$f(d) = d - \frac{2}{3\sqrt{3}}(1-d)^{3/2} + O\big((1-d)^2\big).$$

The same formula holds for $f^*(d)$.

Proof. For fixed $d < 1$ sufficiently close to one, graphs in the definition have $m/n^2 = (1-d)/2 + O(1/n)$. Theorem 1.3 bounds $D_3(G)$ from above, and Proposition 2.3 with $r = 3$ gives a matching lower bound at exactly the required integer edge count. Divide their uncovered-edge bounds by $\binom{n}{2}$, and let n tend to infinity. The leading coefficient becomes $2c_3/2^{3/2} = 2/(3\sqrt{3})$, and the remainder is $O((1-d)^2)$. This proves both statements. □

3 Stability

We extract the structure from the packing constructed in the proof of Theorem 1.2. Near equality first determines the division of missing edges between the exceptional vertices and the core. Turán stability then identifies a common multipartite set inside the uncovered neighborhoods. A local improvement of a maximum clique packing shows that the whole neighborhoods are close to this set. The integer and fractional arguments differ only in the final packing of the dense core. Separating that contribution gives the same structural estimates throughout the superlinear range.

Write $t_p(u)$ for the number of edges in the balanced complete p-partite graph on u vertices. We use the following known quantitative form of Turán stability. If a K_{p+1}-free graph J on u vertices has $e(J) = t_p(u) - a$, then some complete p-partite graph on $V(J)$ differs from J in at most $3a$ edges [5, Corollary 2]. For $p = 1$, the required conclusion is simply that a K_2-free graph has no edges.

Proof of Theorem 1.6. We prove both assertions using the same exceptional-vertex construction. For integer packings put $\sigma = D_r(G) - pn/2$, and for fractional packings put $\sigma = D_r^*(G)$. In either case the hypothesis is $\sigma \geq (c_r - o(1))m^{3/2}/n$. Write $q = m/n^2$, $L = m^{3/2}/n$, and $S_0 = \sqrt{2pm/3}$. Apply the construction in the proof of Theorem 1.2. In this proof denote its original core by K and its exceptional set by E, to distinguish them from the partition we seek. With $b = |E|$, that construction gives $b \leq 2q^{3/4}n = o(\sqrt{m})$. During the exceptional-vertex stages, the graph of edges used inside the core has maximum degree at most $pb = o(\sqrt{m})$.

Retain the set Z from that construction. For $x \in Z$, let J_x be the current core when x is processed, let $W_x = N_G(x) \cap K$, and let U_x be the vertices left uncovered by the maximum vertex-disjoint K_{p+1}-packing chosen in $J_x[W_x]$. Put $z = |Z|$, $u_x = |U_x|$, and let M be the number of missing edges in the original graph $G[K]$. In particular, $J_x[U_x]$ is K_{p+1}-free. The shared budget and the uncovered-edge estimate in that proof give, uniformly in x,

$$\frac{zn}{m} + \frac{M}{m} \leq 1 + o(1), \qquad u_x \leq \sqrt{2pM} + o(\sqrt{m}), \qquad \sigma \leq \sum_{x \in Z} u_x + o(L). \tag{15}$$

For integer packings, Lemma 2.1 bounds the final core loss by $pn/2 + O_r(1)$. For fractional packings, decompose the same core fractionally using [2, Theorem 1.3], and give weight one to all previously selected cliques. Their edge supports are disjoint. In both cases the remaining error is $O_r(q^{7/4}n^2 + q^{3/4}n + 1) = o(L)$ for $n \ll m \ll n^2$. These estimates concern the actual packing construction; no structural assumption about G is being made.

Claim 4. *The parameters satisfy $z = (2/3 + o(1))m/n$ and $M = (1/3 + o(1))m$. There is a set $C \subseteq Z$ with $|C| = (1 - o(1))z$ such that $u_x = (1 + o(1))S_0$ uniformly for $x \in C$.*

Proof. Put $a = zn/m$ and $d = M/m$. The hypothesis and (15) imply $a\sqrt{d} \geq 2/(3\sqrt{3}) - o(1)$ and $a + d \leq 1 + o(1)$. On the triangle $a, d \geq 0$, $a + d \leq 1$, the function $a\sqrt{d}$ has the unique maximum $2/(3\sqrt{3})$ at $(a, d) = (2/3, 1/3)$. Compactness gives the asserted values of z and M.

The upper bound for u_x is now $S_0 + o(S_0)$ uniformly, while (15) gives $\sum_{x \in Z} u_x = (1 + o(1))zS_0$. Choose $e_n \to 0$ such that $u_x \leq (1 + e_n)S_0$ for all $x \in Z$. The nonnegative deficits $(1 + e_n)S_0 - u_x$ have sum $o(zS_0)$. Markov's inequality, with a threshold tending to zero sufficiently slowly, gives C and the stated uniform estimate. □

Claim 5. *There is a set $B \subseteq K$ with a partition $B = B_1 \cup \cdots \cup B_p$ such that $|B_i| = S_0/p + o(\sqrt{m})$, the graph $G[B]$ differs from the corresponding complete p-partite graph in $o(m)$ edges, and $\max_{x \in C} |U_x \triangle B| = o(\sqrt{m})$. Furthermore, all but $o(m)$ missing edges of $G[K]$ lie within the sets B_i.*

Proof. Choose $x_0 \in C$ and put $B = U_{x_0}$. The number of missing edges of $J_{x_0}[B]$ is at most $M + pb|B|/2 = M + o(m)$. Turán's theorem bounds it below by $|B|^2/(2p) - |B|/2 = M - o(m)$. Thus $e(J_{x_0}[B]) = t_p(|B|) - o(m)$. The quoted stability theorem supplies a complete p-partite graph differing from $J_{x_0}[B]$ in $o(m)$ edges. Passing back to $G[B]$ changes at most $pb|B|/2 = o(m)$ edges. The sizes of its parts satisfy $\sum_i (|B_i| - |B|/p)^2 = o(m)$, by comparing its edge count with $t_p(|B|)$. Hence $|B_i| = S_0/p + o(\sqrt{m})$. The missing edges of $G[B]$ number $M - o(m)$, so only $o(m)$ original core missing edges lie outside B; only $o(m)$ of those inside B lie between distinct parts.

For every $x \in C$, the same Turán estimate on $J_x[U_x]$ shows that $G[U_x]$ contains $M - o(m)$ missing edges, uniformly in x. Indeed, its order is $(1 + o(1))S_0$ uniformly, and at most $pb|U_x|/2 = o(m)$ of the current missing edges were used by earlier cliques. Consequently,

$$\binom{|B_i|}{2} - \binom{|B_i \cap U_x|}{2} = o(m)$$

uniformly for every $x \in C$ and i. The possible edges inside $G[B_i]$ contribute only $o(m)$ to this comparison. Since $|B_i| \sim S_0/p$, it follows that $|B_i \setminus U_x| = o(\sqrt{m})$ uniformly. The estimates $|U_x| = (1 + o(1))S_0$ and $|B| = (1 + o(1))S_0$ then give the symmetric-difference assertion. □

The graph of used core edges can have more than $o(m)$ edges in total. The preceding argument only used its maximum degree on sets of order $O(\sqrt{m})$. The same degree bound suffices for the local improvement that follows.

Claim 6. *The neighborhoods satisfy* $\max_{x \in C} |N_G(x) \triangle B| = o(\sqrt{m})$.

Proof. Choose $\tau = \tau_n \to 0$ sufficiently slowly, and define

$$E_1 = \{v \in K \setminus B : |B \setminus N_G(v)| \geq \tau S_0\}.$$

Let $E_2 \subseteq B$ consist of vertices with at least τS_0 nonneighbors in the other parts of B. Claim 5 implies $|E_1|, |E_2| = o(\sqrt{m})$ if τ tends to zero slowly enough. This follows by dividing the $o(m)$ missing edges in the relevant pairs by τS_0. Choose the same τ to dominate the normalized used-edge maximum degree pb/S_0 and the uniform alignment error of that claim.

Fix $x \in C$, and let $\mathcal{P}_x$ be the maximum vertex-disjoint K_{p+1}-packing chosen in $J_x[W_x]$. Each set $V_i = U_x \cap (B_i \setminus E_2)$ has size $S_0/p - o(\sqrt{m})$, uniformly in x. In J_x, a vertex of $(K \setminus B) \setminus E_1$ has only $o(\sqrt{m})$ nonneighbors in B, and a vertex of $B \setminus E_2$ has only $o(\sqrt{m})$ nonneighbors in the other parts. Both statements include the at most pb used edges per vertex.

Suppose that a member Q of $\mathcal{P}_x$ avoids $E_1 \cup (B \setminus U_x)$. Since Q is disjoint from U_x, all its vertices lie in $(K \setminus B) \setminus E_1$. For each of its $p + 1$ vertices a, form a new K_{p+1} using a and one vertex from each V_i. These choices can be made greedily and disjointly. Each V_i has order $S_0/p - o(\sqrt{m})$, adjacency to the bounded number of previously chosen vertices excludes only $o(\sqrt{m})$ choices, and earlier new cliques occupy only $O_r(1)$ vertices. Replacing Q by these $p + 1$ cliques increases the packing size. The replacement uses only Q and uncovered vertices, so it does not meet any other member of $\mathcal{P}_x$. This contradicts the choice of $\mathcal{P}_x$.

Every member of $\mathcal{P}_x$ therefore meets $E_1 \cup (B \setminus U_x)$, a set of size $o(\sqrt{m})$ uniformly in x. The members are vertex-disjoint, so $|W_x \setminus U_x| = (p + 1)|\mathcal{P}_x| = o(\sqrt{m})$. By Claim 5, $|W_x \triangle B| = o(\sqrt{m})$ uniformly. There are only $b = o(\sqrt{m})$ vertices outside the original core, which proves the claim. □

Set $A = V(G) \setminus (B \cup C)$ and form H as in the theorem. Since $N_H(x) = B$ for every $x \in C$, Claim 6 bounds the number of erroneous pairs incident with C by $\sum_{x \in C} |N_G(x) \triangle B| = o(|C|\sqrt{m}) = o(L)$. The number of original missing edges incident with C is

$$|C|n + O\left(|C|^2 + \sum_{x \in C} d_G(x)\right) = \left(\frac{2}{3} + o(1)\right) m.$$

Here $|C|^2 = o(m)$ and the degree sum is $O(L) = o(m)$. The missing edges inside B number $(1/3 + o(1))m$, and these two classes are disjoint. Thus all remaining missing edges, including those inside A and between A and B, number $o(m)$. Together with the $o(m)$ errors inside B from Claim 5, this gives $|E(G) \triangle E(H)| = o(m)$ and completes the proof. □

4 Concluding remarks

Theorems 1.1 and 1.2 determine the first loss from a complete packing near the complete-graph endpoint. They do not determine the extremal packing density at an arbitrary fixed density. Even for triangles, the intermediate-density part of the problem in [3] remains beyond these estimates.

The two error scales in Theorem 1.6 are different for a necessary reason. Near extremality alone does not force the entire graph to be within $o(m^{3/2}/n)$ edits of the construction. The following example shows that unrelated missing edges in the large clique can exceed the packing-loss scale.

Fix r and take the construction from Proposition 2.3 with $m_0 \sim n^{3/2}$. Choose disjoint sets $X, Y \subseteq A$ of size $t = \lfloor n^{11/16} \rfloor$ and delete all their joining edges. The $O_r(n)$ deletions used to prescribe m_0 can be made outside $X \cup Y$, so the complement on $X \cup Y$ is exactly $K_{t,t}$. The resulting graph has $m = m_0 + t^2 \sim m_0$ missing edges. Its original BC edges still lie in no K_r, so $D_r^*(G) \geq (c_r - o(1))m^{3/2}/n$. Theorem 1.2 gives the matching upper bound for $D_r(G)$, since here $n = o(m^{3/2}/n)$.

Consider any three-part template H as in Theorem 1.6, with $|C_H| = O(m/n)$. Removing C_H from X, Y leaves sets X', Y' of orders $a, b = t - o(t)$, since $|C_H| = O(n^{1/2}) = o(t)$. The complement of H outside C_H is a union of vertex-disjoint cliques and isolated vertices. If its parts on $X' \cup Y'$ contain a_i and b_i vertices from the two sets, respectively, then their edit distance from $K_{a,b}$ is

$$ab - \sum_i a_i b_i + \sum_i \left(\binom{a_i}{2} + \binom{b_i}{2} \right) = ab - \frac{a+b}{2} + \frac{1}{2} \sum_i (a_i - b_i)^2 \geq (1 - o(1))t^2.$$

Thus $|E(G) \triangle E(H)| \geq (1 - o(1))t^2$, whereas $t^2 \sim n^{11/8} \gg m^{3/2}/n \sim n^{5/4}$. A bound at the smaller scale for all edges would require a quantitative assumption on the rate of approach to extremality.

The remainder for $r \geq 4$ may also admit an improvement. The proof of Theorem 1.2 uses only that the uncovered part of a maximum K_{r-1}-packing is K_{r-1}-free. For triangles, short augmenting paths supply the stronger inequality (13), which leads to the quadratic error. An analogous estimate for larger clique packings could reduce the error in (1).

Theorem 1.1 separates the worst-case degree-residue loss from the structural loss throughout $n \ll m \ll n^2$. The transition at $m \asymp n^{4/3}$ is therefore included, although the formula does not describe the loss of each individual graph. For fractional packings, Theorem 1.4 determines the leading loss at every linear defect $m/n \to \lambda$, including its changes as the number of exceptional vertices increases. When $\lambda \leq 1$, the theorem gives only an $o(\sqrt{n})$ loss; a finer description would complement the exact small-defect result of Gruslys and Letzter [8].

For integer packings at linear defect, Theorem 1.5 determines the residue-dependent linear term and the structural square-root term. The latter has different thresholds from the fractional profile, because the largest core degree residues consume part of the missing-edge budget. The remaining questions include a finer error on the positive branches and a structural description of their near-extremal graphs. The $O_r(1)$ estimate on the unsaturated branch also leaves open the exact bounded correction, which may depend on further congruences.

Declaration of generative AI use

OpenAI Codex was used to explore candidate arguments, examine proof steps, locate related literature, and revise the organization and wording of the manuscript. Its use during the research is described at the beginning of Section 2.

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